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Lab 2: Agile Backlog Creation and Sprint Simulation in Jira

Problem Statement 16: Remote Patient Vitals Alert and Monitoring App

1. What I built

I took the five functional requirements and two non functional requirements from Lab 1 and grouped them into four Epics. Under those I wrote ten User Stories in the As a / I want / So that format, gave each one a priority and a story point estimate from the Fibonacci scale, and ran two one week sprints on the Jira board.

The whole backlog came to 58 points across ten stories. Sprint 1 took the five High priority stories, which was 26 points. Sprint 2 took the five Medium priority stories plus three stories that did not finish in Sprint 1, which came to 48 points.

Epic	Lab 1 requirements covered	Stories	Points
Vitals Collection and Monitoring	FR-001, NFR-001	3	16
Limit Checking and Alerts	FR-002, FR-005	3	16
Caregiver Escalation	FR-003	2	13
Settings and Security	FR-004, NFR-002	2	13
Total		**10**	**58**

2. Reflection questions

Did your estimations reflect the actual effort?

I simulated the sprints rather than writing any code, so there are no real hours to compare the estimates against. The relative sizing still holds up when I look at it again. The 8 point stories are the ones that either depend on another system or have to keep working under load: handling 500 patients streaming at once, raising an alert inside a 2 second window, passing an alert down the caregiver list on a timeout, and encrypting readings while restricting who can see them. The 3 point ones are single, narrow behaviours, like the emergency button or a warning when a device stops sending. What I could not test here is whether an 8 really costs roughly twice a 5, and that only comes from comparing estimates against time actually spent.

Was your backlog well-prioritized?

Mostly yes. The five High stories in Sprint 1 cover the path that has to exist before anything else is worth building: readings arrive, they get checked against the safe limits, an alert is raised, a caregiver is notified, and the next caregiver is tried if the first one is not reached. Without those the app does nothing useful for a post operative patient at home. The Medium stories are all things that make the system better to live with rather than things it cannot run without.

One thing I would change. I left the story about raising an alert within 2 seconds at Medium priority, so it fell into Sprint 2. That requirement was High priority in my Lab 1 table and it is the core of the whole problem statement, so it belonged in Sprint 1. That was an inconsistency between my requirements table and my backlog rather than a deliberate call.

How did your simulated sprint align with your plan?

Sprint 1 did not go to plan. I committed 26 points across five stories and only closed two of them, worth 10 points. The other three, worth 16 points, were still in the To Do column when I completed the sprint, so Jira pushed them back to the backlog. Sprint 2 then started at 48 points because it picked up those three stories on top of the five Medium ones, and it closed everything.

That is a real carryover rather than a tidy simulation, and it matches what the lab notes say about overcommitting during sprint planning. The plan assumed a full sprint of work could be moved through in one sitting, and it could not. Sprint 2 cleared because it had a longer run at it, not because those stories were easier.

What insights did the burndown chart give about your team's capacity?

The Sprint 1 chart drops from 26 to 16 in a single step near the start and then stays flat for the rest of the week, which is what an overcommitted sprint looks like. The remaining line sits well above the grey guideline from that point on, and on a real project that gap is the signal to stop adding work and cut scope instead. The Sprint 2 chart falls all the way to zero, so the whole 48 points cleared.

Velocity across the two sprints was 10 and then 48. That is far too uneven to plan a third sprint against. A useful velocity number needs several sprints of a similar shape before it means anything, and with one person and a simulated week neither figure is trustworthy on its own. If I had to pick a starting point I would use something closer to the average of the two than to either number by itself.

3. Screenshots

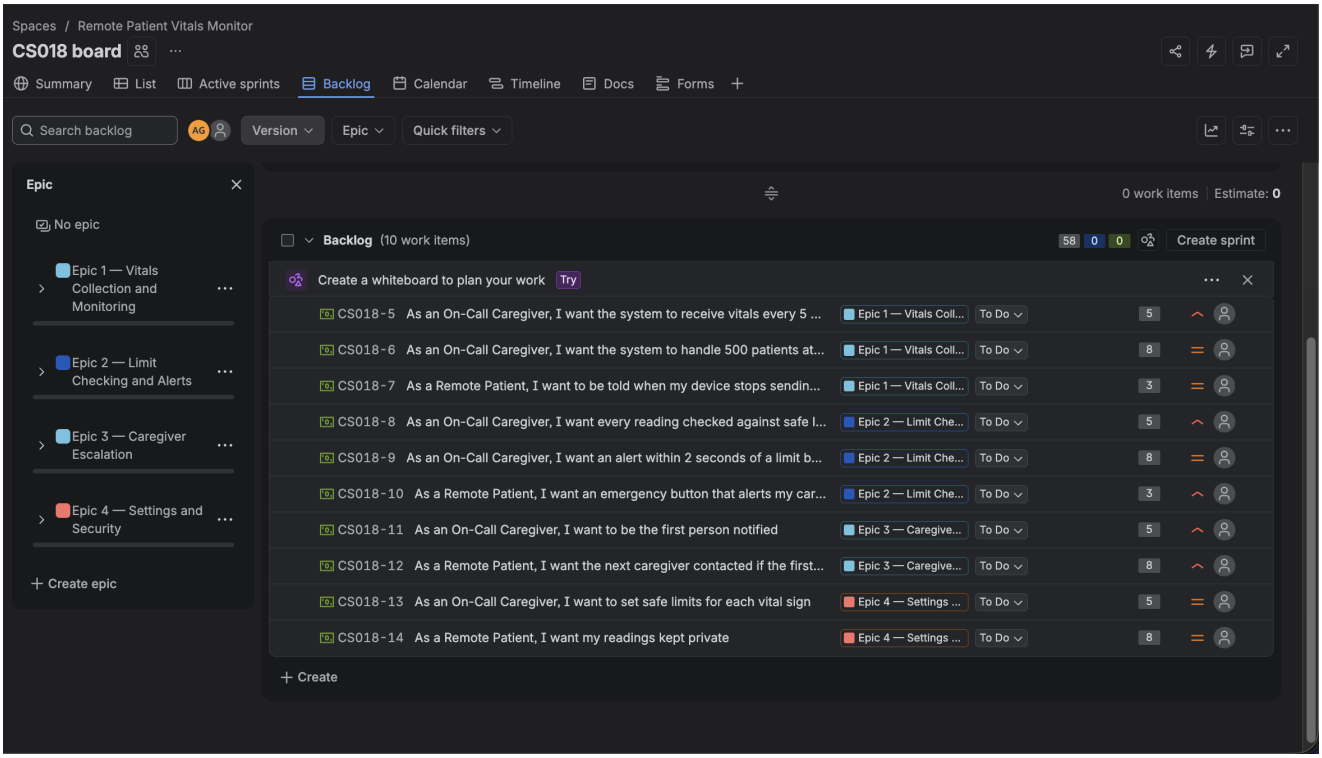


Figure 1. Product backlog with the four Epics and ten User Stories, each tagged to its Epic and carrying a priority and a story point estimate. Total 58 points.

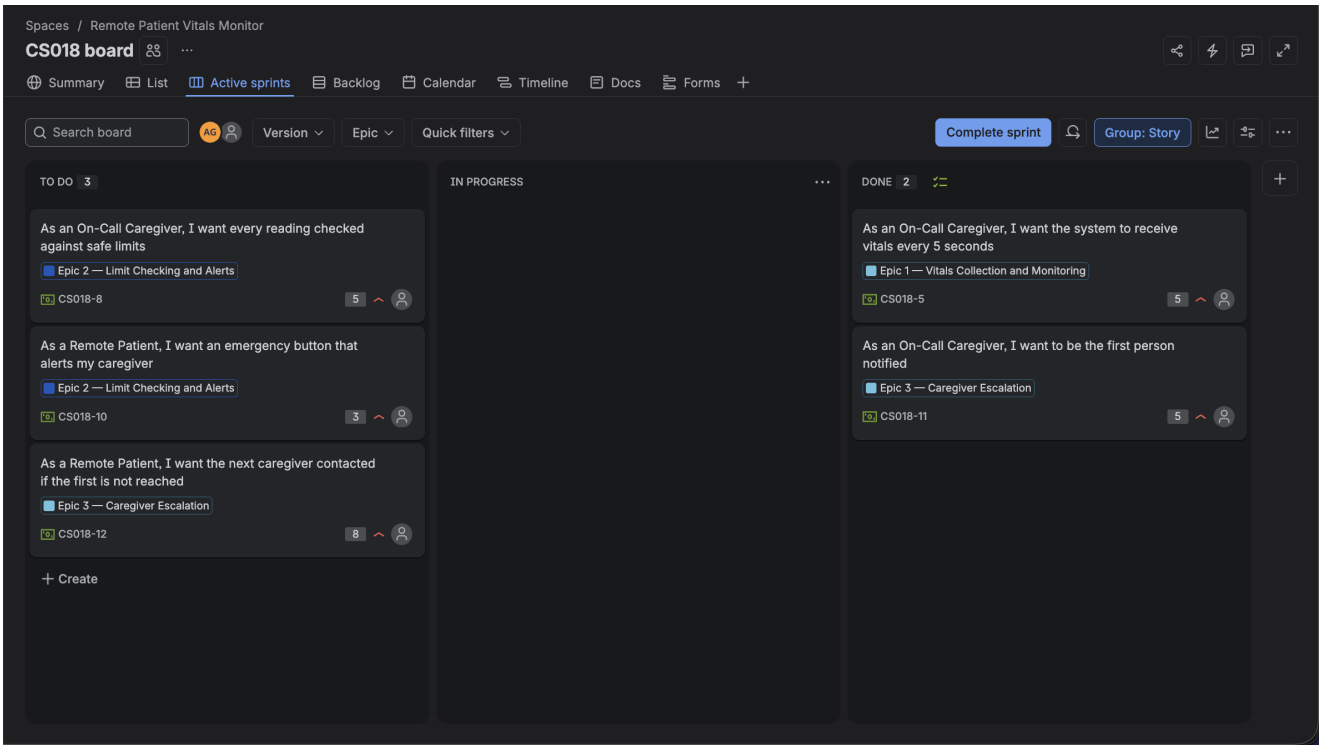


Figure 2. Active sprint board for Sprint 1, part way through, with two stories moved to Done and three still in To Do.

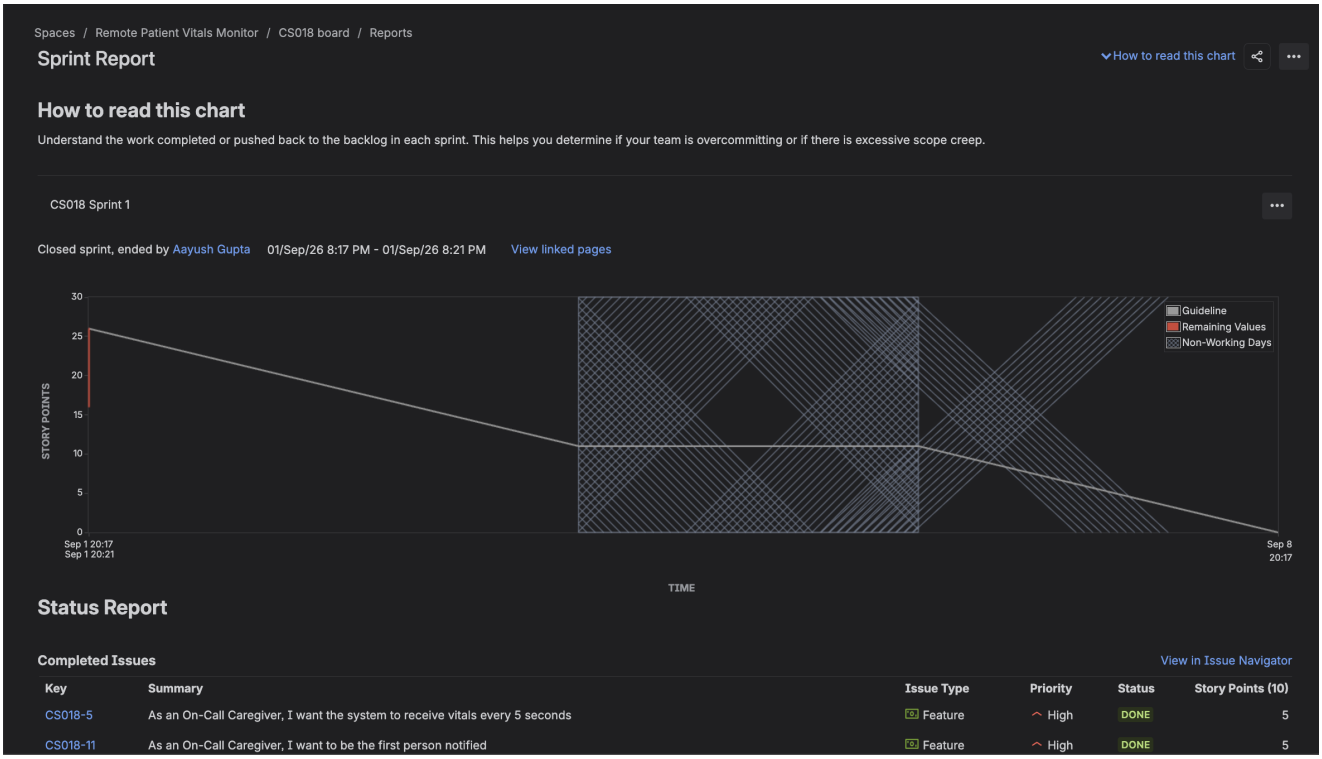


Figure 3. Sprint 1 report. Committed 26 story points, completed 10, with 16 points pushed back to the backlog.

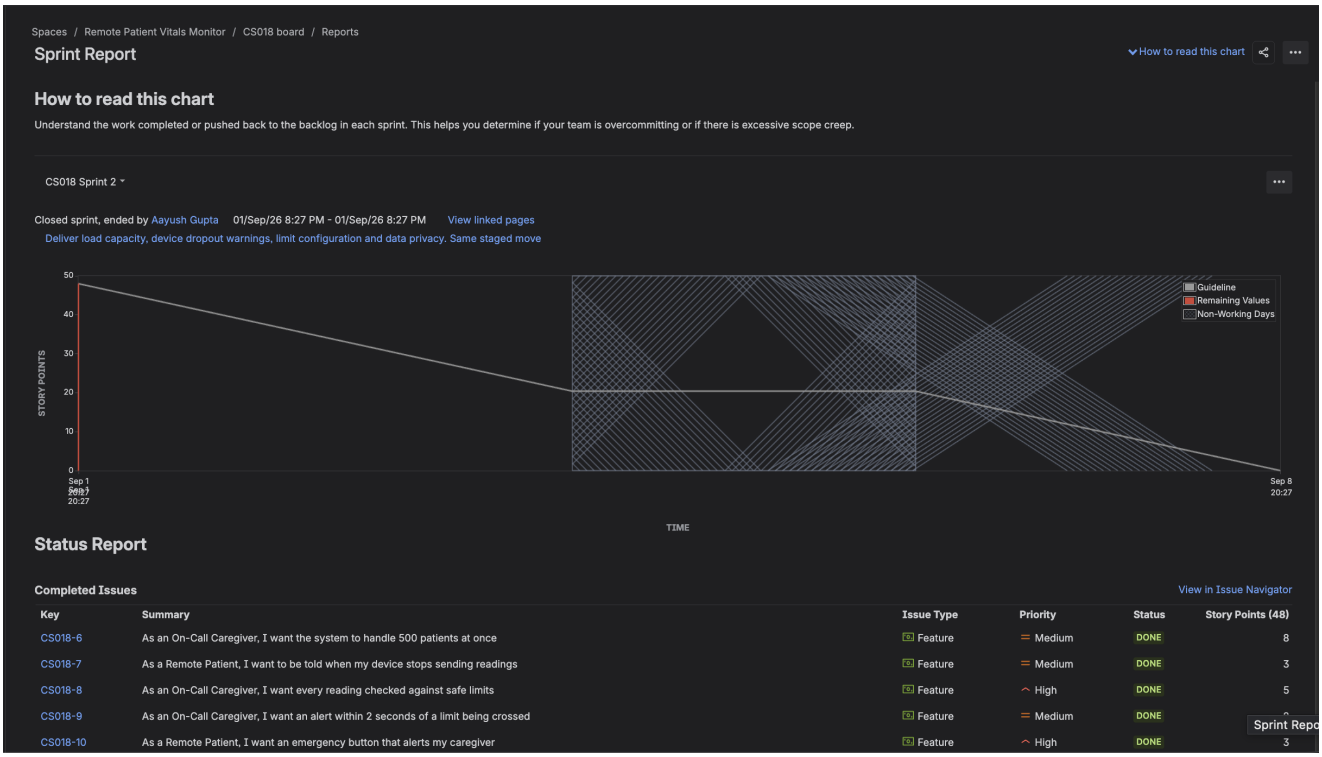


Figure 4. Sprint 2 report. Started at 48 story points, made up of the five Medium stories plus the three carried over from Sprint 1, and closed all of them.